

6809 Flex..... - stardot.org.uk

 stardot.org.uk/forums/viewtopic.php

Prime

June 17, 2019

[6809 Flex.....](#)

[Prime](#)

Posts: 3149

Joined: Mon Jun 01, 2009 12:52 am

[6809 Flex.....](#)

[Quote](#)

Hi all,

I think I remember chatting to someone at one of the meetups that had got Flex working on the Co-Pro, but had compiled under a modern OS.

Could that person remind me how they did it, I have a disk with the Flex sources on... I need to try and integrate some disk drivers into the kernel, and thought now might be a good time to port the Flex sources to a modern 6809 assembler, but would be good to start with something known to work.....

Cheers.

Phill.

[Top](#)

 [User](#)
[avatar](#)

[sydney](#)

Posts: 3021

Joined: Wed May 18, 2005 10:09 am

Location: Newcastle upon Tyne

Re: 6809 Flex.....

[Quote](#)

[Post](#) by [sydney](#) »

It could well have been [JimRussell](#). He's not been on for a while but he's on the Facebook group as James Russell so you might have more luck finding him there.

[Top 100 games list as voted by you!](#)

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 [User](#)
[avatar](#)

[danielj](#)

Posts: 10648

Joined: Thu Oct 02, 2008 5:51 pm

Location: Manchester

Re: 6809 Flex.....

[Quote](#)

[Post](#) by [danielj](#) »

RobC got flex running on the copro too iirc?

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[Prime](#)

Posts: 3149

Joined: Mon Jun 01, 2009 12:52 am

Re: 6809 Flex.....

[Quote](#)

| [danielj](#) wrote: Mon Jun 17, 2019 3:01 pm RobC got flex running on the copro too iirc?

Yeah that rings bells.

Cheers.

Phill.

[Top](#)

[roland](#)

Posts: 5334

Joined: Thu Aug 29, 2013 9:29 pm

Location: Born (NL)

Contact:

[Contact roland](#)

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [roland](#) »

It could be me. I adapted Flex for my 6809 copro. I added drivers for console and disk I/O. The 6809 send commands to the Atom (non tube) and the Atom executes it.

IIRC there is no boot disk in this setup. Flex is loaded from the boot routine; it simply asks the Atom to transfer the Flex OS file(s) to the 6809. That stores the files in its memory and then boots Flex. From then on it will use disk images.

I developed and assembled the Flex files on an OS X machine.

Goodbye Misty ...

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 [User](#)
[avatar](#)

[dominicbeesley](#)

Posts: 2681

Joined: Tue Apr 30, 2013 12:16 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [dominicbeesley](#) »

There's a thread here: [viewtopic.php?t=9114&start=120](#) where RobC helped me get it running on an LX9/matchbox copro.

I also ported it to my own 6809 [viewtopic.php?f=3&t=14125&start=30#p202781](#) I have to admit my memory isn't great at the best of times but I've forgotten most of this but if you need help I can do some digging.

I think I used the Xroar assemblers but I really can't remember. I got a wd1770 native to my 6809 beeb going but didn't do much with it

D

[Top](#)

 [User](#)
[avatar](#)

[Elminster](#)

Posts: 4341

Joined: Wed Jun 20, 2012 9:09 am

Location: Essex, UK

Contact:

[Contact Elminster](#)

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Elminster](#) »

I also remember talking to RobC about Flex on my 6800 SWTPC. So he could well be the person.

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [RobC](#) »

Hi Phill,

I am at work at the moment but will dig out my notes etc. this evening and, if you PM me your email address, I'll mail you the details.

I got the sources from Jim Davis in the Flex User Group and, after writing my own drivers, compiled Flex using a 6809 compiler on Windows (think it was a09 but I might be mistaken). If you haven't seen it already, the FLEX Adaptation Guide is really useful.

Best wishes,

Rob

[Top](#)

[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

Rob,

I'd be interested in what you have too. I am aware of [JGH's Flex page](#) which, as far as I can see, seems to contain the extra bits that are needed to make Flex work on a 6809 2nd processor but presumably these must be combined with the core OS somehow?

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

Re: 6809 Flex.....

[Quote](#)

[Post](#) by **RobC** »

I think JGH did his own set of drivers after I'd written mine.

I can put what I've got on Dropbox if you want. Just PM me and I'll send you a link.

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

Re: 6809 Flex.....

[Quote](#)

[Post](#) by **Coeus** »

I did try the BBCFLEX09.DSD from the other thread but that doesn't seem to boot in the new B-Em 6809 co-pro. I did start looking at it, though. It looks like the first two sectors are very clever in that they seem to contain the boot loader, as per the Flex documentation, and yet at the same time a valid DFS catalogue. That catalogue seems to contain two files both of which consist only of these two sectors, presumably so Shift-Break and *FLEX both load these boot sectors.

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

Re: 6809 Flex.....

[Quote](#)

[Post](#) by **RobC** »

The clever dual format catalogue came from JGH'.

Did you try running it with *FLEX or SHIFT-Break?

I don't think I ever got SHIFT-Break working

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

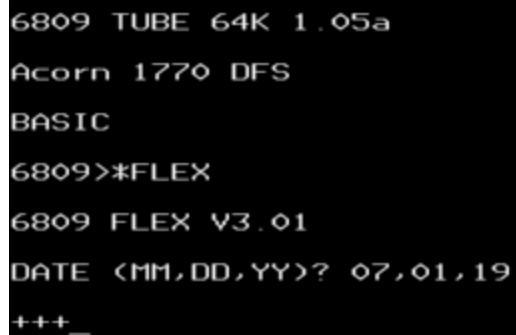
[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

[RobC](#) wrote: Mon Jul 01, 2019 7:55 pm Did you try running it with *FLEX or SHIFT-Break?

I was trying Shift-Break. *FLEX does work:



```
6809 TUBE 64K 1.05a
Acorn 1770 DFS
BASIC
6809>*FLEX
6809 FLEX V3.01
DATE (MM,DD,YY)? 07,01,19
+++_
```

and

```

PRINT      .CMD      2
PRINT      .SYS      1
PROT       .CMD      1
QCHECK     .CMD      4
RELASMB    .CMD     64
RENAME     .CMD      1
RENUMBER   .CMD      2
RENUMX     .CMD      2
SAVE       .CMD      2
SAVE       .LOW      2
SPLIT      .CMD      1
TOTAL2     .INC      7
TOUCH      .CMD      1
TTYSET     .CMD      2
VER        .CMD      3
VERIFY     .CMD      1
VERSION    .CMD      1
XBASIC     .CMD     79
XOUT       .CMD      2
Y          .CMD      1
ZAP        .CMD      3

SECTORS LEFT = 814

+++

```

[RobC](#) wrote: Mon Jul 01, 2019 7:55 pm The clever dual format catalogue came from JGH'.

But isn't this, from your BBCFLEX09.DSD image doing the same:

```

6809 TUBE 64K 1.05a
Acorn 1770 DFS
BASIC
6809>*EX
$. !BOOT      L  00C100 00C120 0000F2 000
$. FLEX       L  00C100 00C120 0000F2 000
6809>* _

```

The two files both seem to have zero as their disk start address. I'll see if I can work out what is going on with !BOOT.

Last edited by [Coeus](#) on Mon Jul 01, 2019 9:06 pm, edited 1 time in total.

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [RobC](#) »

[Coeus](#) wrote: Mon Jul 01, 2019 9:05 pm But isn't this, from your BBCFLEX09.DSD image doing the same:

...

The two files both seem to have zero as their disk start address. I'll see if I can work out what is going on with !BOOT.

Yes - I couldn't work out why SHIFT-Break wouldn't work as it should do the same as *FLEX

Glad it's working for you though. I've uploaded the sources, documents, utilities and DSK images I got to Dropbox in case you want to have a play with stuff yourself.

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

[Coeus](#) wrote: Mon Jul 01, 2019 9:05 pm I'll see if I can work out what is going on with !BOOT.

Rob, this loop here doesn't seem to have a terminating condition:

Code: [Select all](#)

C159 8D 35	BSR	\$C190
C15B 81 02	CMPA	#\$02
C15D 27 10	BEQ	\$C16F
C15F 81 16	CMPA	#\$16
C161 26 F6	BNE	\$C159
C163 8D 2B	BSR	\$C190

C165 B7 C1 47	STA	\$C147
C168 8D 26	BSR	\$C190
C16A B7 C1 48	STA	\$C148
C16D 20 EA	BRA	\$C159
C16F 8D 1F	BSR	\$C190
C171 B7 C1 49	STA	\$C149
C174 8D 1A	BSR	\$C190
C176 B7 C1 4A	STA	\$C14A
C179 8D 15	BSR	\$C190
C17B 1F 89	TFR	A,B
C17D 4D	TSTA	
C17E 27 D9	BEQ	\$C159
C180 BE C1 49	LDX	\$C149
C183 34 14	PSHS	X,B
C185 8D 09	BSR	\$C190
C187 35 14	PULS	B,X
C189 A7 80	STA	,X+
C18B 5A	DECB	
C18C 26 F5	BNE	\$C183
C18E 20 C9	BRA	\$C159

That seems to be a loop over the Flex binary record types looking at types \$02 and \$16 but when it sees \$16 it doesn't then jump to the transfer address, it carries on and, when I ran it in the debugger, it is looping through "records" with a record type of zero, or more likely just padding at the end of the file.

Assuming this is the same file for both !BOOT and *FLEX cases that doesn't explain why *FLEX works, but maybe why !BOOT doesn't.

EDIT: That turns out to be a red herring. It does exit from the subroutine that loads the next sector, i.e. it is checking EOF by looking at the sector links rather than the record types.

Last edited by [Coeus](#) on Mon Jul 01, 2019 9:45 pm, edited 2 times in total.

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

[RobC](#) wrote: Mon Jul 01, 2019 9:30 pm Glad it's working for you though. I've uploaded the sources, documents, utilities and DSK images I got to Dropbox in case you want to have a play with stuff yourself.

That will be useful. Did you send me a link? There's nothing in my Inbox.
Last edited by [Coeus](#) on Mon Jul 01, 2019 9:37 pm, edited 1 time in total.

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [RobC](#) »

[Coeus](#) wrote: Mon Jul 01, 2019 9:32 pm Rob, this loop here doesn't seem to have a terminating condition:

That code comes straight from the Bootstrap loader in the Flex Adaptation guide. I think it's meant to drop out of the get character routine at \$C190.

[Coeus](#) wrote: Mon Jul 01, 2019 9:33 pm That will be useful. Did you send me a link?
There's nothing in my Inbox.

I was in the middle of PMing you when I replied. Should be there now.

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

[RobC](#) wrote: Mon Jul 01, 2019 9:57 pm That code comes straight from the Bootstrap loader in the Flex Adaptation guide. I think it's meant to drop out of the get character routine at \$C190.

Yes, I did find that in the end.

[RobC](#) wrote: Mon Jul 01, 2019 9:57 pm I was in the middle of PMing you when I replied.
Should be there now.

Found the link now. Very useful.

Last edited by [Coeus](#) on Mon Jul 01, 2019 11:03 pm, edited 1 time in total.

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

[RobC](#) wrote: Mon Jul 01, 2019 9:30 pm Yes - I couldn't work out why SHIFT-Break
wouldn't work as it should do the same as *FLEX

Strangely, it turns out it has not crashed but is waiting for keyboard input. Having released Shift-Break, if you then type a character the Flex banner and prompt then appear:



```
6809 TUBE 64K 1.05a
Acorn 1770 DFS
6809 FLEX V3.01
DATE (MM,DD,YY)? 07,01,19
+++_
```

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

And I do believe I have cracked it. Before printing the banner, Flex calls CRLF1 which calls ESCTST which does:

Code: [Select all](#)

```
*TEST KEYBOARD FOR ESCAPE
CE90 BDCD4E      ESCTST JSR STAT character ready?
CE93 2765        BEQ CRLF5 branch if not
CE95 AD9FD3E5    JSR [INCHNE] input w/o echo
CE99 B1CC0A      CMPA TTYESC is it escape?
CE9C 265C        BNE CRLF5 branch if not
CE9E 7FCC1A      ESCTS1 CLR CURLIN clear line count
CEA1 AD9FD3E5    ESCTS2 JSR [INCHNE] look for character
```

So what is happening is when you hit the Break key the OS places the code \$DC in the keyboard buffer which is the softkey code for break (*KEY 10) . This would be expanded when OSRDCH is called. The STAT routine is using OSBYTE \$80 to check the keyboard buffer and finds the \$DC token so then ESCTST calls INCHNE which in turn calls OSRDCH which goes to read the character. At that point OSRDCH on the host side concludes \$DC expands to the empty string so then waits for a fresh character to enter the keyboard buffer from a real keypress.

This can be tested. After power-on set *KEY 10 to a single character, for example:

```
6809 TUBE 64K 1.05a
Virtual DFS
BASIC
6809>*KEY 10 !
6809>*
```

then do Shift-Break and:

```
6809 TUBE 64K 1.05a
Acorn 1770 DFS

6809 FLEX V3.01
DATE (MM,DD,YY)? _
```

with no need to hit any keys.

As for a neater way around it, one possibility would be to have the bootloader or an init hook in one of the driver modules clear the keyboard buffer before Flex gets round to the first ESC check. That would mean no softkey expansion for the break key but that's probably the lesser of the two evils.

Last edited by [Coeus](#) on Tue Jul 02, 2019 12:49 am, edited 1 time in total.

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [RobC](#) »

Well done

Now you mention it, I'd noticed the key press thing (or someone else told me about it) but had forgotten it.

[Coeus](#) wrote: Tue Jul 02, 2019 12:22 am As for a neater way around it, one possibility would be to have the bootloader or an init hook in one of the driver modules clear the keyboard buffer before Flex gets round to the first ESC check. That would mean no softkey expansion for the break key but that's probably the lesser of the two evils.

Agree it's the lesser of two evils - I'll have a think where to put it.

[Top](#)

[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....System Information Record \(SIR\)](#)

[Quote](#)

[Post](#) by [Coeus](#) »

I have found the Flex adaptation guide to be slightly misleading in the section that deals with the system information record. The adaptation guide on page 32 dealing with the NEWDISK program gives this table:

Code: [Select all](#)

Begin	End	Information
----	----	-----
0	1	Two bytes of zeroes (Clears forward link)
16	26	Volume name in ASCII
27	28	Volume number in binary
29	30	Address of first data sector (Track-Sector)

```
31    32 Address of last data sector (Track-Sector)
33    34 Total number of data sectors in binary
35    37 Current date (Month-Day-Year) in binary
38    38 Highest track number on disk in binary
39    39 Highest sector number on a track in binary
```

For the entries from offset 29 through to 34, whilst it is true that the NEWDISK program should set these to values for the non-bad data sectors on the disk, FLEX actually uses these to manage the free chain thus these would be better described as:

Code: [Select all](#)

```
29    30 Address of first free data sector (Track-Sector)
31    32 Address of last free data sector (Track-Sector)
33    34 Total number of free data sectors in binary
```

I had been initially puzzled that the values in these fields on the BBCFLEX09.DSD file in Rob's download in the linked thread above did not look like the start of track 1 and the end of track 79 and the total number of sectors didn't look like 80 tracks of 20 sectors.

Then it seemed odd that there should be nowhere else in the SIR documented as holding the head of the free chain. It would also be expected to cache the amount of free space somewhere to avoid having to traverse the whole chain to find it.

Finally, suspecting that these may be the free sector chain I checked the count against the value reported by FLEX itself for this disc, 814, and it matches.

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 [User](#)
[avatar](#)

[jgharston](#)

Posts: 6467

Joined: Thu Sep 24, 2009 12:22 pm

Location: Whitby/Sheffield

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [jgharston](#) »

I think that's what I worked out as well:

Code: [Select all](#)


```

d_num%=d_sec%:d_bps%=256:d_sec0%=1
REM          drv trk sid sec num
A%=FNfdc_disk(&53,mem%,0,0,0,3,1):REM Load System Info Record
d_free%=256*mem%?&21+mem%?&22
d_trk%=mem%?&26
d_sec%=mem%?&27
IF d_sec%>19:d_sid%=2 ELSE d_sid%=1
d_num%=d_sec% DIV d_sid%
d_size% =(d_trk%+1)*d_sec%
d_used%=d_size%-d_free%
mem%?&1B=13:d_title$=FNs($(mem%+&10))
d_cmn%=mem%?&23
d_cdy%=mem%?&24
d_cyr%=mem%?&25+1900
PRINT"System Information Record for ";d_title$;" "
PRINT"Disk shape: ";FNd(d_trk%+1,5);" tracks          ";FNd(d_sec%,3);" sectors
";d_sid%;" sides"
PRINT"Disk used: ";FNd(d_used%,5);" sectors ";FNd(d_used%*256,8);" bytes
";FNd(d_used%/4,5);"K"
PRINT"Disk free: ";FNd(d_free%,5);" sectors ";FNd(d_free%*256,8);" bytes
";FNd(d_free%/4,5);"K"
PRINT"Disk size: ";FNd(d_size%,5);" sectors ";FNd(d_size%*256,8);" bytes
";FNd(d_size%/4,5);"K"
PRINT"Created:          ";FNd0(d_cdy%,2);"/";FNd0(d_cmn%,2);"/";d_cyr%

```

Code: [Select all](#)

```

$ bbcbasic
PDP11 BBC BASIC IV Version 0.45
(C) Copyright J.G.Harston 1989,2005-2024
>_

```

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[paulxtr](#)

Posts: 20

Joined: Sun Apr 15, 2012 5:02 pm

Location: Manchester, UK

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [paulxtr](#) »

Any chance of fixing the error reporting in this version of FLEX?

For example if you try to

DELETE FOO.BAR

you get the following error (assuming the file doesn't exist):

DISK ERROR #4

when you should get the appropriate message from the ERRORS.SYS file:

THE SPECIFIED FILE COULD NOT BE FOUND

I noticed that the File Sector Map Indicator byte is not set to \$02 (random access file) in the ERRORS.SYS file, but the error reporting still doesn't work even when it is set to \$02.

Cheers, Paul

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[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

[paulxtr](#) wrote: *Wed Jul 03, 2019 6:47 pm* Any chance of fixing the error reporting in this version of FLEX?

Try this one: What I did was mount the original TSCFLX09.DSK as drive 1 (requires to link/rename/copy it to have .ssd extension in b-em) then with the bbcflex9.dsd as drive 0 and FLEX booted:

Code: [Select all](#)

DELETE ERRORS.SYS
COPY,1,ERRORS.SYS,0

I am guessing that as well as the random access flag in the directory it may have lost the FSM in a previous copy.

Last edited by [Coeus](#) on Wed Jul 03, 2019 9:53 pm, edited 3 times in total.

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[paulxtr](#)

Posts: 20

Joined: Sun Apr 15, 2012 5:02 pm

Location: Manchester, UK

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [paulxtr](#) »

[Coeus](#) wrote: Wed Jul 03, 2019 9:45 pm

[paulxtr](#) wrote: Wed Jul 03, 2019 6:47 pm Any chance of fixing the error reporting in this version of FLEX?

Try this one:

bbcflex9.zip

Yes, it's reporting errors properly now.

Excellent work, thanks.

Cheers,

Paul

[Top](#)

[Coeus](#)

Posts: 4094

Joined: Mon Jul 25, 2016 12:05 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [Coeus](#) »

So having got this far has anyone developed anything to run under Flex? If so, what tools did you use.

RobC sent me a cross assembler which seems to be [this one on GitHub](#). That seems to be cross-platform and can produce the Flex-specific binary format which is good. How do people get files to/from Flex discs?

Is there a native assembler? Is that ASMB.CMD is? It doesn't seem to be documented in either the programmers guide, adaptation guide or user's guide.

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[RobC](#)

Posts: 4223

Joined: Sat Sep 01, 2007 10:41 pm

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [RobC](#) »

[Coeus](#) wrote: Thu Jul 04, 2019 4:05 pm So having got this far has anyone developed anything to run under Flex? If so, what tools did you use.

I've written some small utilities (MODE, FCOL, BCOL) that allow Flex to access the Beeb's modes and colours.

[Coeus](#) wrote: Thu Jul 04, 2019 4:05 pm

RobC sent me a cross assembler which seems to be [this one on GitHub](#). That seems to be cross-platform and can produce the Flex-specific binary format which is good. How do people get files to/from Flex discs?

I use the Windows FloppyMaint application which should be in the dump of stuff I sent you. Let me know if it's not and I'll send it to you.

[Coeus](#) wrote: Thu Jul 04, 2019 4:05 pm Is there a native assembler? Is that ASMB.CMD is? It doesn't seem to be documented in either the programmers guide, adaptation guide or user's guide.

The manual (along with many others!) is [here](#).

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[jgharston](#)

Posts: 6467

Joined: Thu Sep 24, 2009 12:22 pm

Location: Whitby/Sheffield

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by [jgharston](#) »

I was about to sit down and update the 6809CoPro client code to fix the KBDTST code, then looking at it remembered the problem I originally had.

It's not possible on the BBC to test if a call to OSRDCH will return a character without actually removing a character from the keyboard buffer. It is only possible to test if there is a byte (not a character) in the keyboard buffer, or if there is currently a soft key being expanded. And that doesn't even get into the problems of if OSRDCH is not reading from the keyboard buffer.

That leaves the edge case of a soft key being the top item in the keyboard buffer, but that soft key not being defined. So, ADVAL(-1) says "there's a byte available in the keyboard buffer!", but OSRDCH find "softkey length zero, keep waiting until something to return". You can avoid the waiting forever bit by using INKEY(0), but that removes a character if one is available. You can either test if a character is returnable **and* *fetch* *it**, or test if there is a byte in the keyboard buffer **which* *may* *not* *return* *anything**.

I hit this problem when writing a serial terminal, but in that case using the first option is ok, because the code never needed to do test if a keypress was pending, but asked: is there a keypress pending, **and* *if* *so*, *give* *it* *to* *me**. Which is the INKEY(0) action.

The only way around this is a lot of convoluted "smelly" code, is an exec file open, is it at EOF?, what stream is OSRDCH connected to?, examine the source buffer, if it's serial, is it being expanded?, is this is an unescaped topbit byte, is it defined as a soft key (ok so far.....) and is that soft key defined? (blam!). There is no API to get information about a soft key string,

you have to know what the host is and how the host stores soft keys, and examine the soft keys directly.

So, I ended up resigning to making KBDTEST just do ADVAL(-1) to test for the contents of the keyboard buffer.

I think the best solution for our purposes is to make the BBC Flex loader clear the keyboard buffer.

*there's a proper term, but I can't remember it.

Code: [Select all](#)

```
$ bbcbasic
PDP11 BBC BASIC IV Version 0.45
(C) Copyright J.G.Harston 1989,2005-2024
>_
```

[Top](#)

[BigEd](#)

Posts: 7559

Joined: Sun Jan 24, 2010 10:24 am

Location: West Country

Contact:

[Contact BigEd](#)

[Re: 6809 Flex.....](#)

[Quote](#)

[Post](#) by **[BigEd](#)** »

It would be possible, I think, to hook into OSRDCH (host side) using RDCHV, with a little stateful routine that has a one-byte buffer and a flag to say if it's full. If INKEY(0) can be intercepted too, it can get a byte from the modified OSRDCH and then put it back, using a kind of ungetc() which puts the fetched byte (if there was one) into the buffer and sets the flag.

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